RL-Tuned Synthetic Marine Radar for Sim-to-Real Transfer

Team Name Placeholder

CSCE 775: Deep Reinforcement Learning

I. Introduction

Marine radar perception and simulation-driven autonomy have received decades of attention because robust sensing is central to safe maritime robotics and monitoring. However, despite progress in radar signal processing and synthetic-data pipelines, real radar data remain expensive and sparse, and fixed simulator settings still produce a persistent sim-to-real gap that limits downstream detection and mapping performance.

This proposal studies whether a reinforcement learning policy can adaptively tune synthetic radar generator parameters to maximize real-data validation performance across environment types (lake, river, coast). The central hypothesis is that environment-conditioned simulator control can outperform fixed or random domain randomization and reduce transfer error under unseen routes and weather/clutter conditions.

The approach is grounded in prior work on domain randomization and adaptive sim-to-real transfer [1]–[13]. However, existing methods either rely on fixed/random randomization, non-RL adaptive tuning, or calibration-centric transfer that is not explicitly conditioned on maritime environment type. Our proposal targets this gap with an environment-aware RL outer loop that directly optimizes real-data downstream task performance, as summarized in Fig. 1 and contrasted with prior work in Fig. 2.

II. Approach

We formulate synthetic radar tuning as a bi-level optimization problem. In the inner loop, a downstream model is trained on synthetic radar generated with parameter vector θ. In the outer loop, an RL policy observes the current generator state, environment-type label, and recent real-validation metrics, then outputs updates to θ.

The action space controls key simulator factors including clutter intensity, noise floor, attenuation, speckle, and target reflectivity variation. The reward is defined as real-data task improvement minus realism/stability penalties to discourage unphysical parameter exploitation. We use Soft Actor-Critic (SAC) as the primary outer-loop optimizer due to sample efficiency and continuous-control support, with TD3 as a backup. A non-RL Bayesian optimization tuner is included as a strong baseline.

Relative to existing adaptive randomization methods (SimOpt, BayesSim, BayRn) [3]–[5], our method makes three concrete changes: (1) sequential policy-based control of simulator updates rather than one-step or purely inference-driven adaptation, (2) explicit conditioning on maritime environment type (‘lake‘, ‘river‘, ‘coast‘), and (3) physically bounded action constraints with realism penalties to prevent simulator exploitation. Relative to radar calibration-oriented sim-to-real pipelines [11], the optimization target is not just simulation fidelity but downstream real-task performance under domain holdout.

III. Evaluation

Evaluation uses real marine radar logs with explicit environment labels and strict split protocols by route/location to prevent leakage. We compare: (1) fixed synthetic defaults, (2) uniform random domain randomization, (3) Bayesian optimization tuning, (4) RL tuning without environment conditioning, and (5) RL tuning with conditioning.

Primary metrics are downstream real-data task score and relative sim-to-real gap reduction. Secondary metrics include cross-domain generalization (train on two environment types, test on held-out type), unseen-location robustness, seed variance, and performance under low-SNR/high-clutter stress conditions. We will report both best-case performance and sample-efficiency measured by real-validation calls required to reach target accuracy.

To explicitly support the novelty claims, we add claim-linked evaluations: (a) sequential adaptation benefit by comparing RL vs BayRn under matched real-evaluation budgets [5]; (b) environment semantics benefit by comparing conditioned vs unconditioned RL policies; and (c) plausibility benefit by reporting parameter-bound violations and instability with/without realism regularization. These analyses directly test how our method improves over existing randomization and simulator-adaptation approaches [1]–[4], [6], [7].

Published results already show clear failure modes for non-adaptive baselines (Fig. 3). In real FMCW radar transfer, simulation-only baselines can collapse to near-chance balanced accuracy (about 50% for occupancy and 33% for people-counting), while calibrated adaptation substantially improves performance (about 97% and 72%, respectively) [11]. In synthetic-to-measured SAR transfer, synthetic-only and simple-noise baselines are consistently weaker than stronger adaptation strategies across multiple architectures [10]. These patterns motivate the need for a more structured, environment-aware adaptation controller rather than fixed randomization.

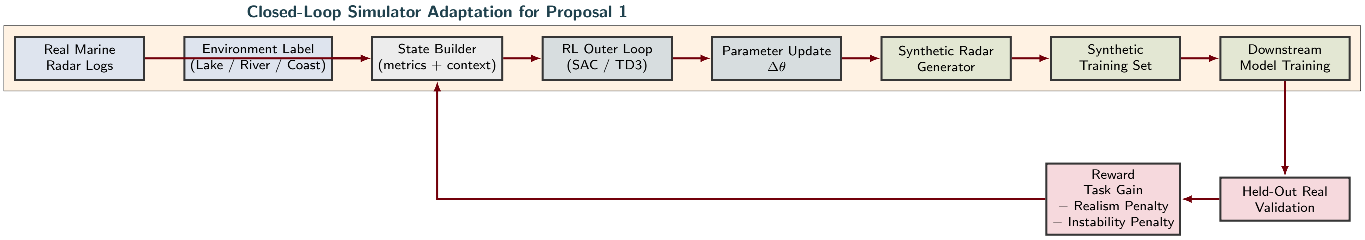

Fig. 1. Proposal 1 closed-loop method: an RL outer loop updates simulator parameters from real-validation reward, conditioned on environment type.

Method Family	Real-Feedback Adaptive	Env-Type Aware	Sequential Policy Control	Physical Bounds	Task-Reward Optimized
Fixed Domain Randomization (Tobin, Peng)	✗	✗	✗	✗	✗
SimOpt / BayesSim	✓	✗	✗	✗	✗
BayRn / ADR / CDR	✓	✗	✗	✗	✗
Radar Calibration (Digital Twin style)	✓	✗	✗	✓	✗
Proposed Method (Proposal 1)	✓	✓	✓	✓	✓

Fig. 2. Related-work capability comparison and the explicit improvement axes targeted by Proposal 1.

IV. Conclusion

This project targets a practical and research-relevant problem: improving sim-to-real transfer for marine radar perception through adaptive synthetic data generation. The expected contribution is an RL-controlled simulator tuning framework that is environment-aware, benchmarked against strong non-RL alternatives, and evaluated under realistic transfer stress tests. If full RL tuning is unstable, a contextual-bandit or Bayesian fallback will still provide a complete, meaningful project outcome.

References

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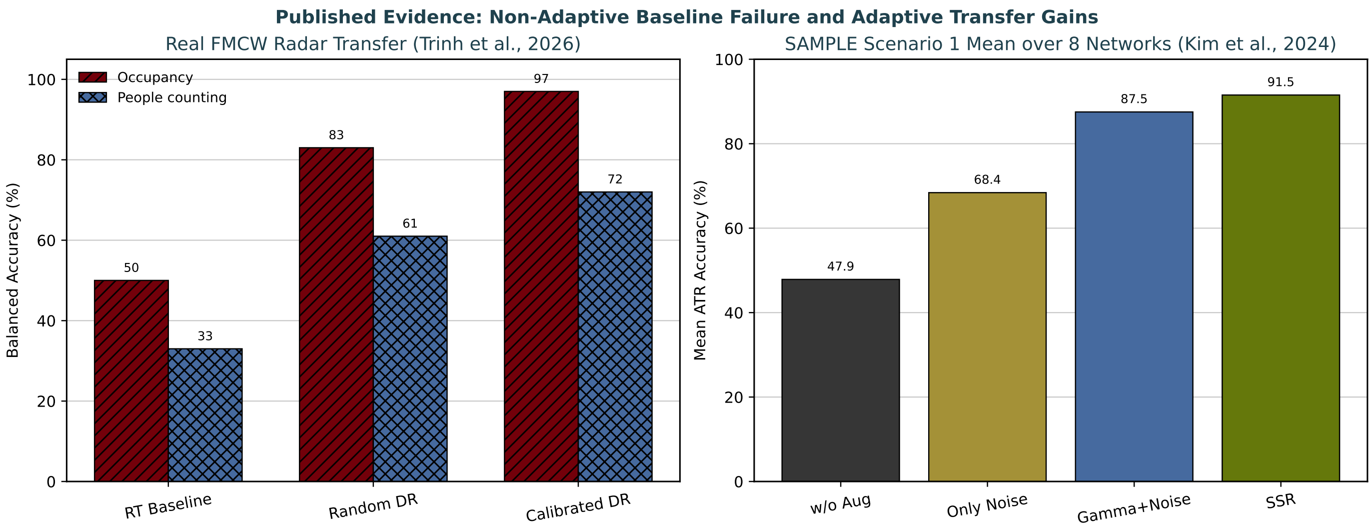

Fig. 3. Published quantitative evidence motivating Proposal 1. Left: non-adaptive and weakly adaptive baselines can fail under sim-to-real radar transfer; calibrated/adaptive methods improve substantially. Right: synthetic-only and simple noise baselines underperform on measured-data transfer compared to stronger adaptation strategies.